



Advanced Conceptual Modeling

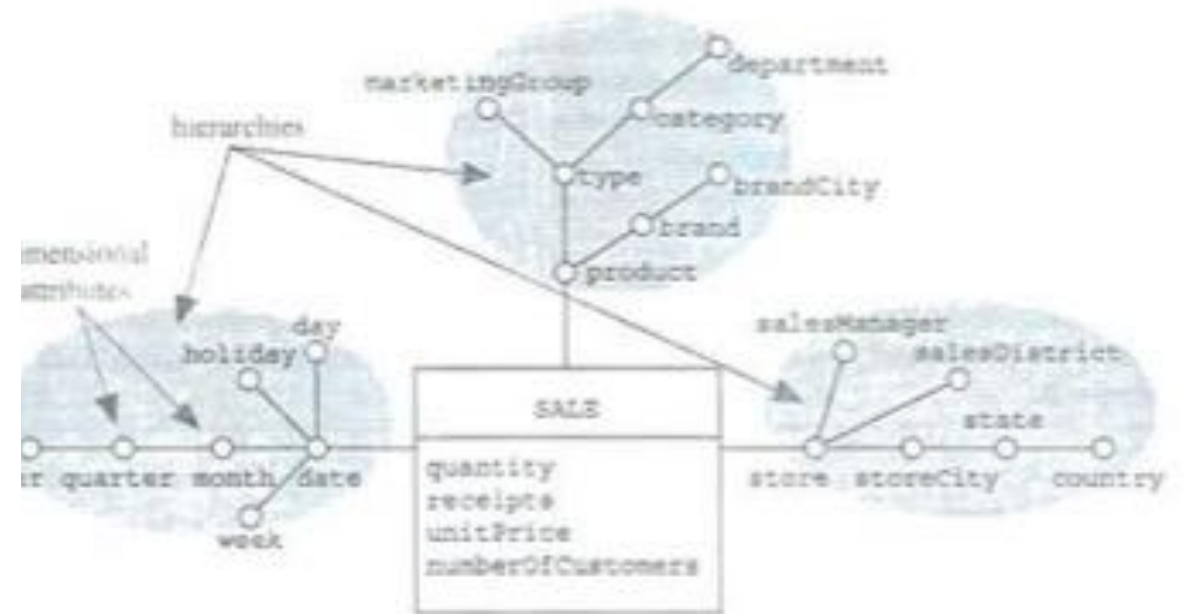
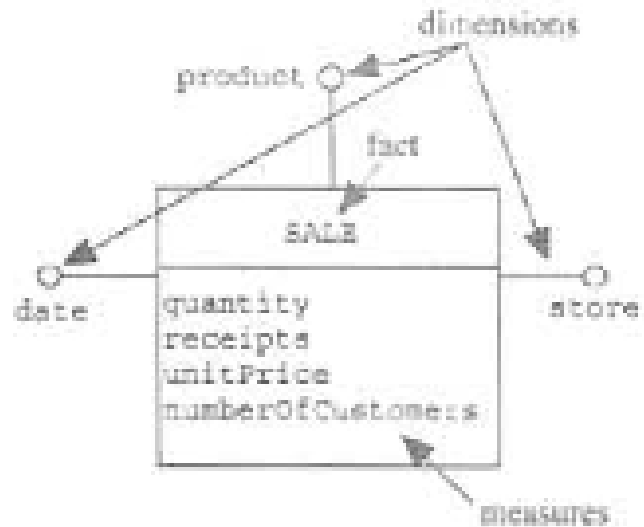
IF5240

INTELIJEN BISNIS & ANALITIK

LIST OF TOPICS

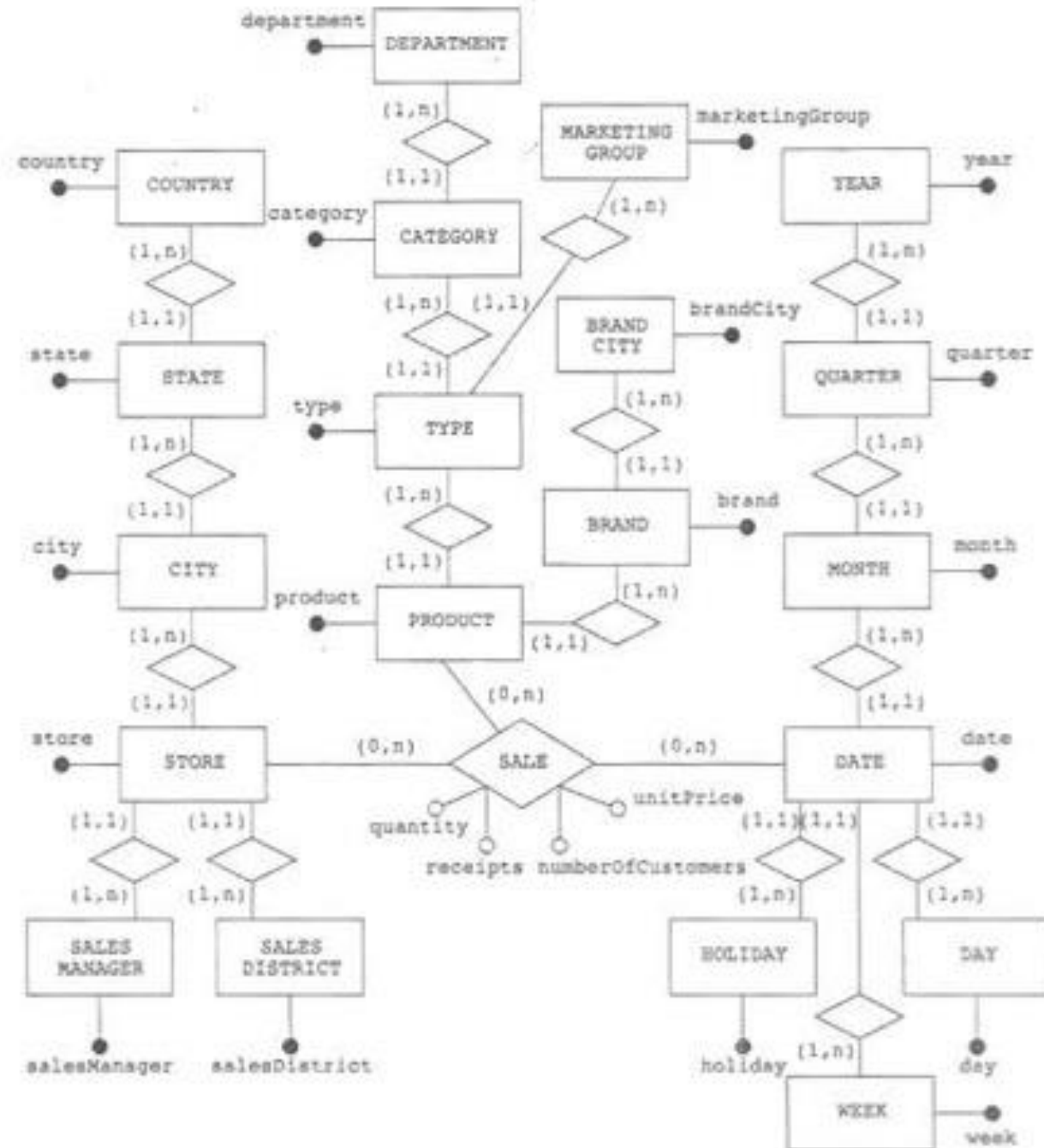
1. Descriptive Attributes
2. Cross Dimensional Attributes
3. Convergence Attributes
4. Shared Hierarchies
5. Multiple Arcs
6. Additivity
7. Aggregation

Review Sales (simple vs enhanced)



Entity-Relationship

FOR ENHANCED SALES FACT SCHEMA



Descriptive Attribute

It is an additional information, which is not used in aggregation criteria

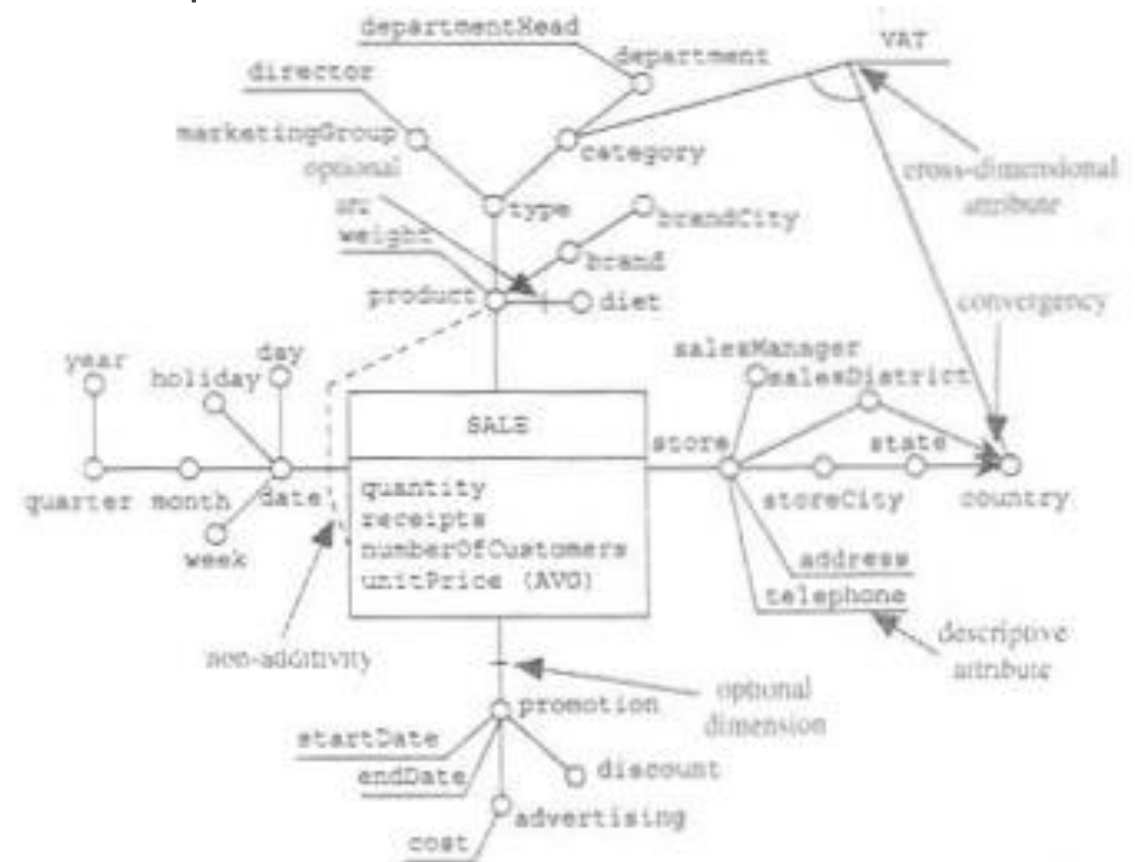
It is determined by one-to-one association

It is usually as “leave” hierarchy

Ex. Store address, telephone number, product weight

Yet, descriptive attribute can also be directly connected with fact

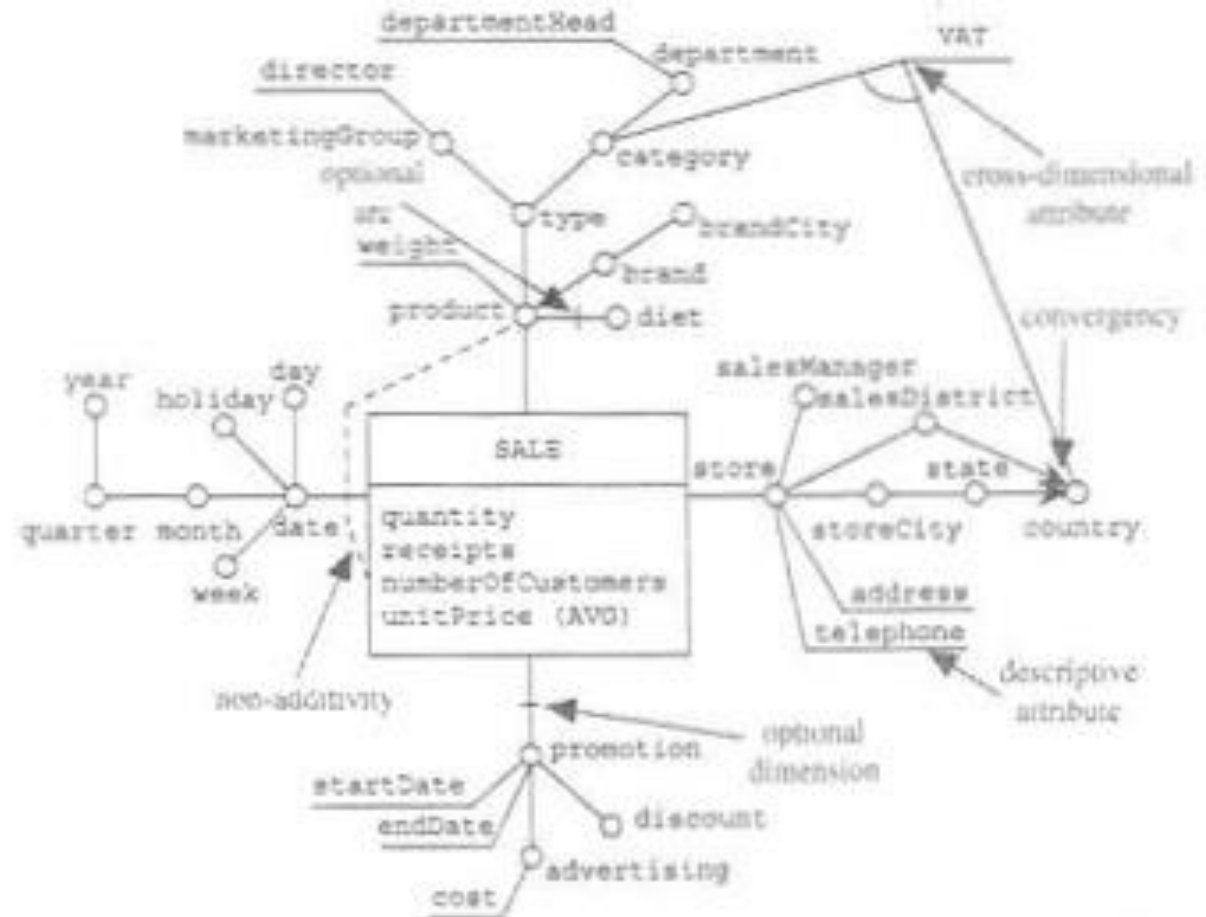
Complete Sales Fact Schema



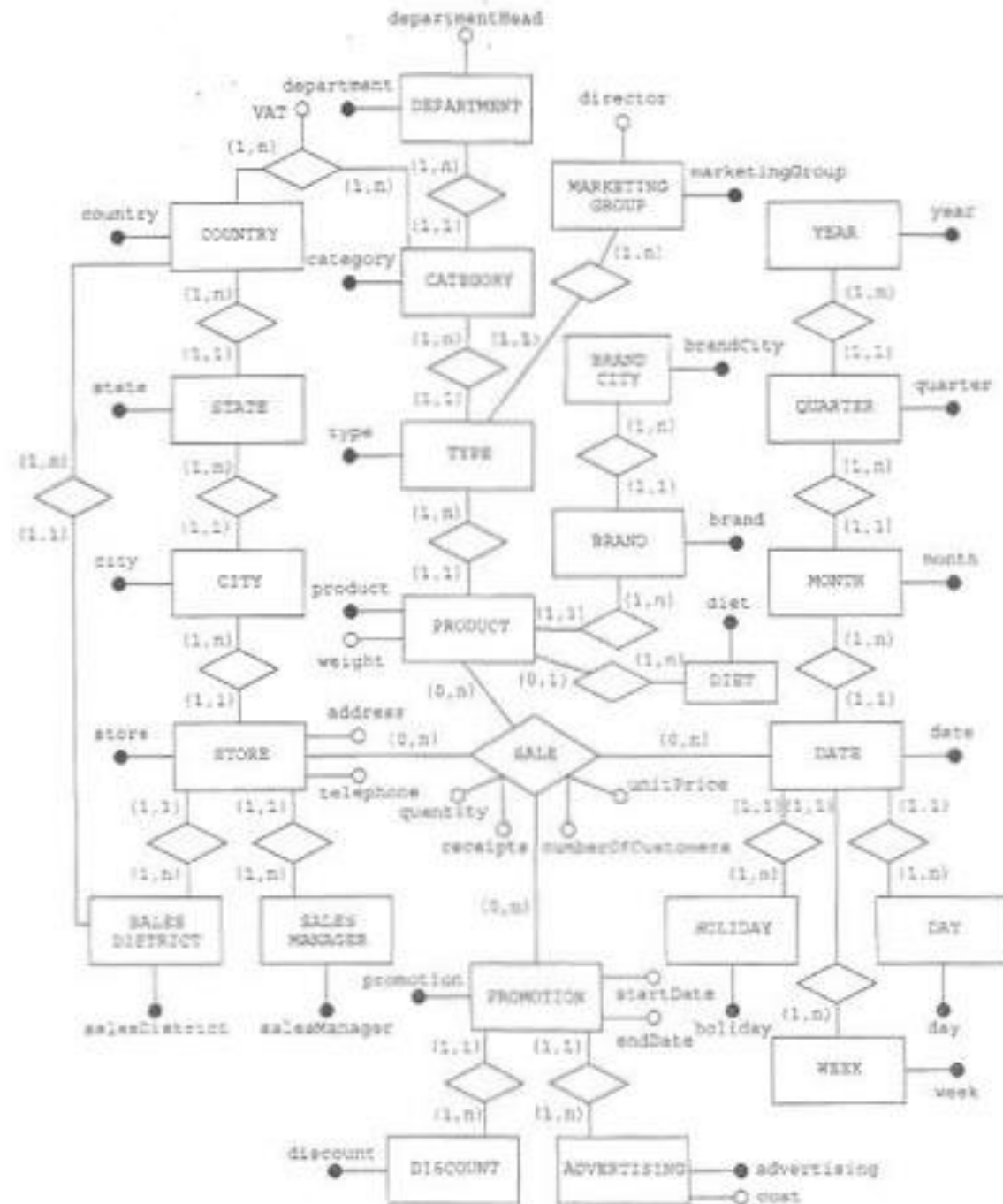
Cross Dimensional Attributes

It is dimensional or descriptive attribute whose value is defined by two or more attributes.

Ex. VAT attribute



Cross Dimensional Attributes



Convergence Attributes

It is structure hierarchies

Ex: store -> storeCity -> state -> country

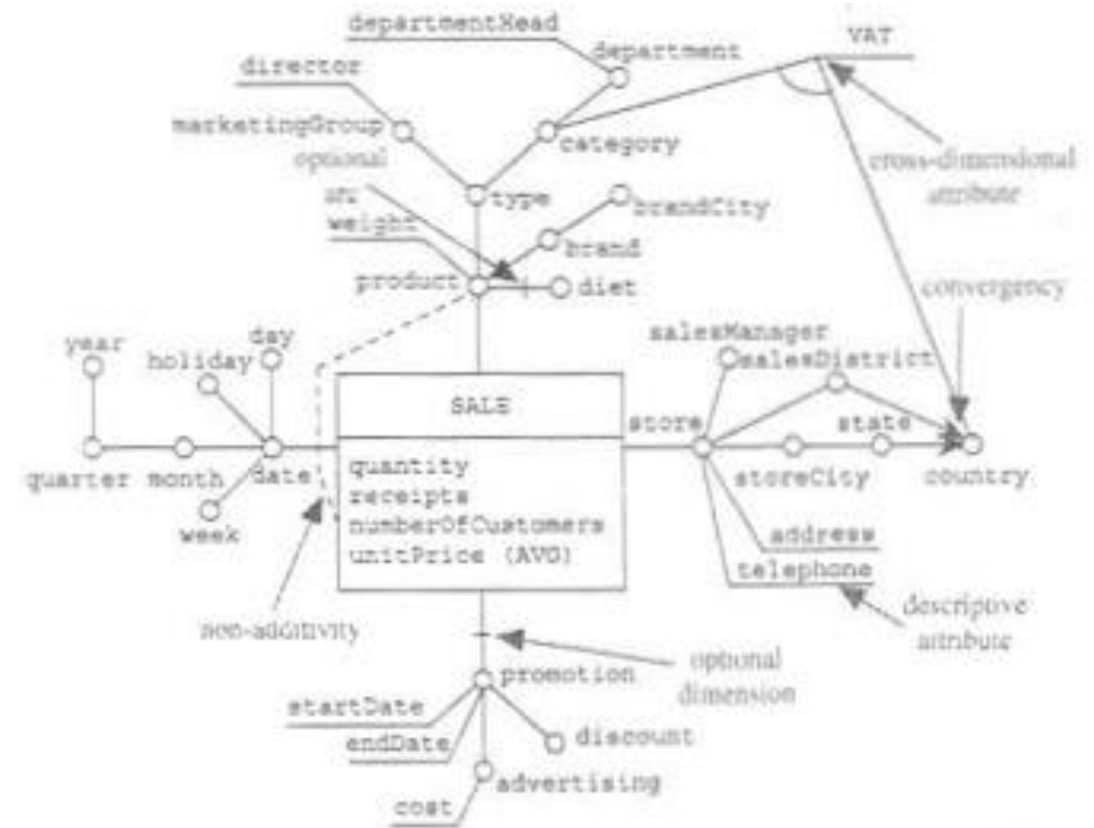
Store -> salesDistrict -> country

It is showed by arrow

How about : Product dimension vs Store Dimension

Brandcity : city where the product is manufactured

StoreCity : city where product is sold

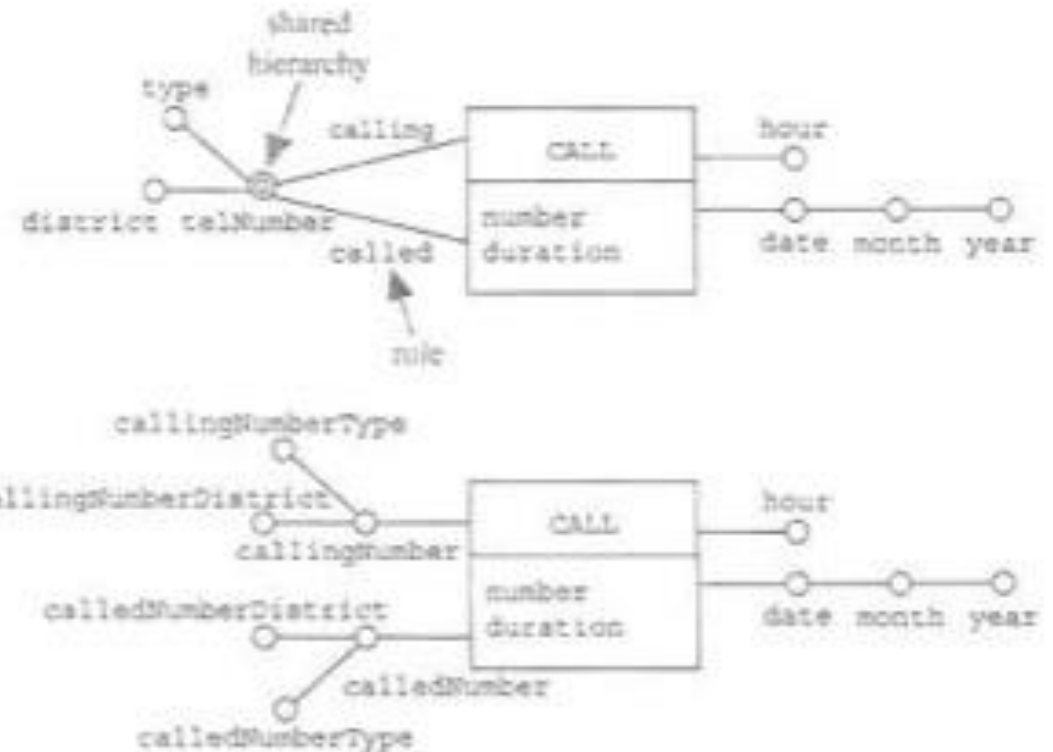


Shared Hierarchies

Hierarchies frequently are replicated in fact schemata. Ex. Temporal hierarchies, geographical hierarchies.

Notation for shared hierarchies is described in phone calls fact schemata, i.e. doubled circle

All descendant attributes are also shared.



Multiple Arcs

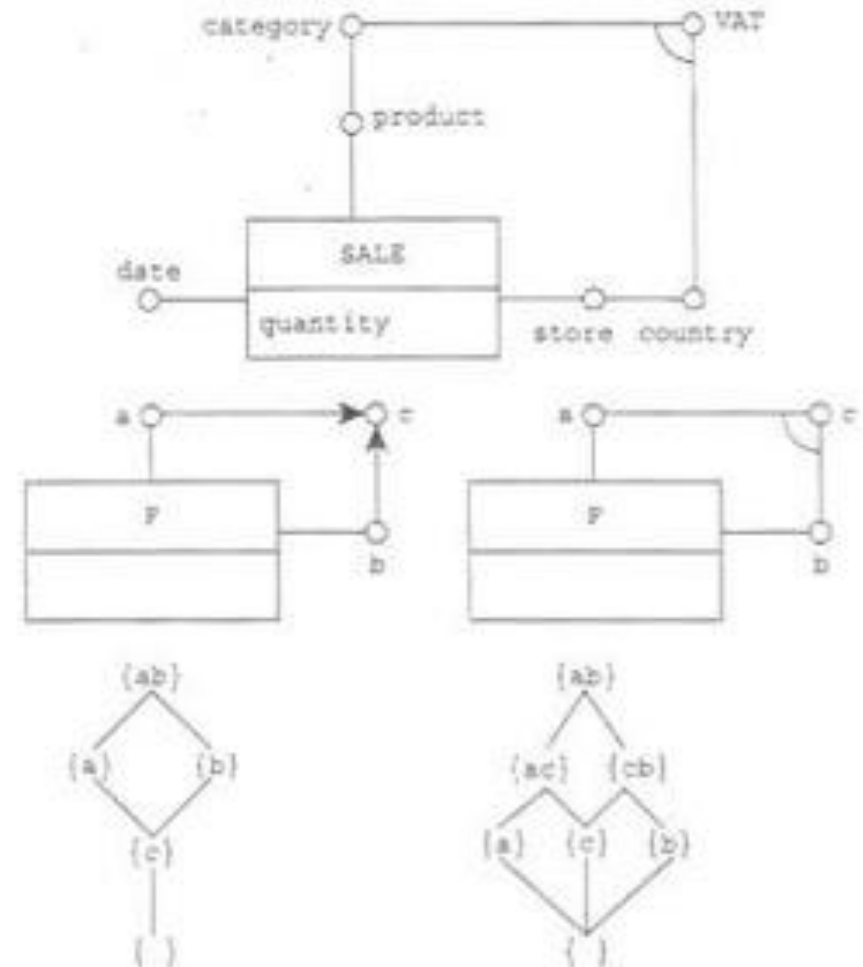
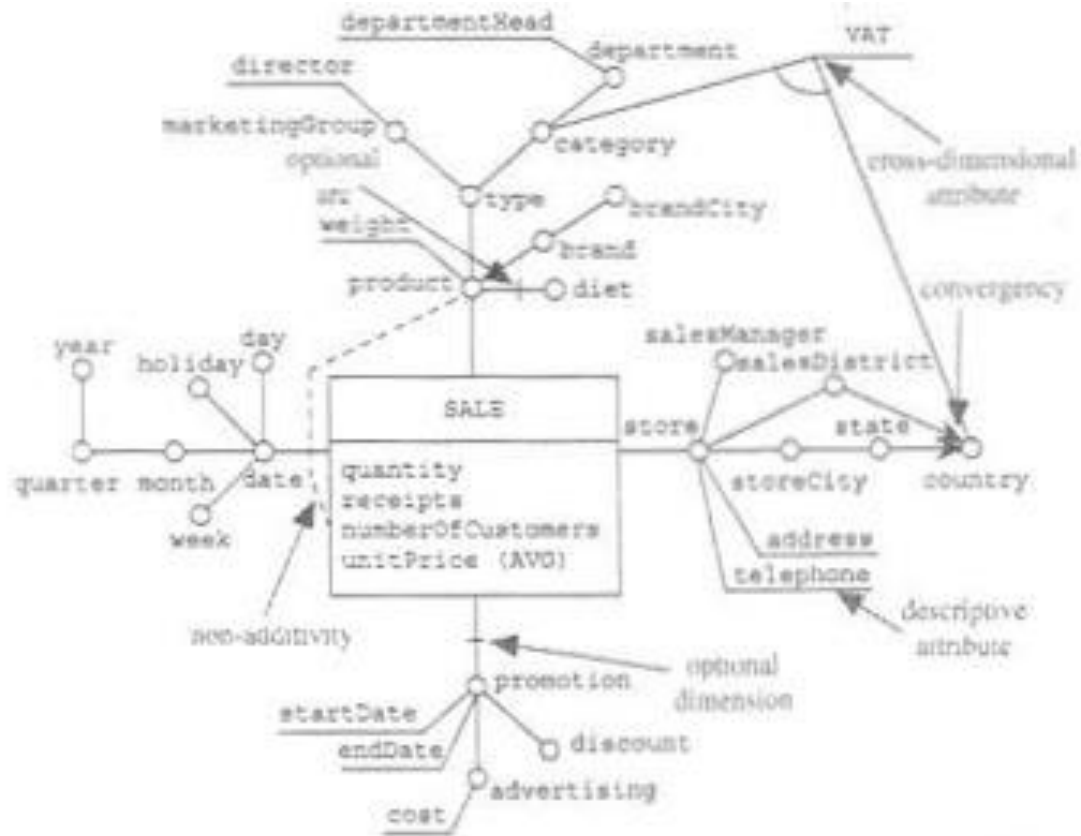
Ex. Fact schemata about book sales in a book store. Common hierarchy is book author.

But there are also some needs to analyze author based on their books. Yet, author can not act as dimensions. (why?)

Notation : double line



Aggregating with Convergence & Cross Dimensional Attributes



Aggregating with Multiple Arcs

genre	author	book	date 1/2/09
Technical	Golfarelli, Rizzi	Facts & Crimes	3
	Golfarelli	Sounds Logical	5
Current affairs	Rizzi	The Right Measure	10
	Golfarelli, Rizzi	Facts, How and Why	4
Science Fiction	Golfarelli	The 4th Dimension	8
Total:			30

TABLE 5-16 Primary Events in the Bookseller Schema

TABLE 5-17 The
(author, date)
Group-by Set
Secondary Events

author	date 1/2/09
Golfarelli	20
Rizzi	17
Total:	37

TABLE 5-18
Multiple Arc
Weights Between
book and author

	Golfarelli	Rizzi
Facts & Crimes	0.5	0.5
Sounds Logical	1	0
The Right Measure	0	1
Facts, How and Why	0.5	0.5
The 4th Dimension	0	0

TABLE 5-19 The
(author, date)
Group-by Set
Weighted
Secondary Events

author	date 1/2/09
Golfarelli	16.5
Rizzi	13.5
Total:	30



Sources

“Datawarehouse Design” : Modern Principles & Methodologies,
M.Golfarelli, S.Rizzi